LOAN APPROVAL PREDICTION USING MACHINE LEARNING

Internship Project Report

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Tools: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Google Colab

1. Introduction

Loan approval is an important decision-making process for financial institutions. Loan applications are evaluated using applicant characteristics such as income, education, employment status, credit history, loan amount, and property area. This project uses supervised machine learning to predict whether a loan application will be Approved or Rejected.

2. Problem Statement

The objective is to develop a machine learning classification model that predicts loan approval based on applicant information, focusing on data cleaning, missing-value handling, categorical encoding, scaling, class imbalance handling, model comparison, evaluation, and business interpretation.

3. Objectives

• Clean and preprocess the dataset.
• Handle missing values.
• Encode categorical variables and scale numerical variables.
• Handle class imbalance using SMOTE.
• Compare Logistic Regression, Decision Tree, and Random Forest.
• Evaluate Precision, Recall, F1-score, and ROC-AUC.
• Perform threshold analysis.
• Provide business insights.

4. Dataset Description

The dataset contains 614 loan applications. Loan_Status is the target, where Y represents Approved and N represents Rejected. Loan_ID was removed because it is an identifier.

5. Data Preprocessing

Categorical missing values were handled using the most frequent category, while numerical missing values were handled using median imputation. Categorical variables were encoded using One-Hot Encoding and numerical variables were standardized using StandardScaler. The data was split into 80% training and 20% testing data using stratification.

6. Handling Class Imbalance

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data within the modeling pipeline to address class imbalance while avoiding information leakage.

7. Machine Learning Models

Logistic Regression was used as a baseline model. Decision Tree was used to model non-linear relationships. Random Forest was used as an ensemble tree-based model.

8. Model Evaluation

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	82.11%	86.21%	88.24%	87.21%	87.59%
Decision Tree	75.61%	80.22%	85.88%	82.95%	70.05%
Random Forest	78.05%	82.22%	87.06%	84.57%	81.52%

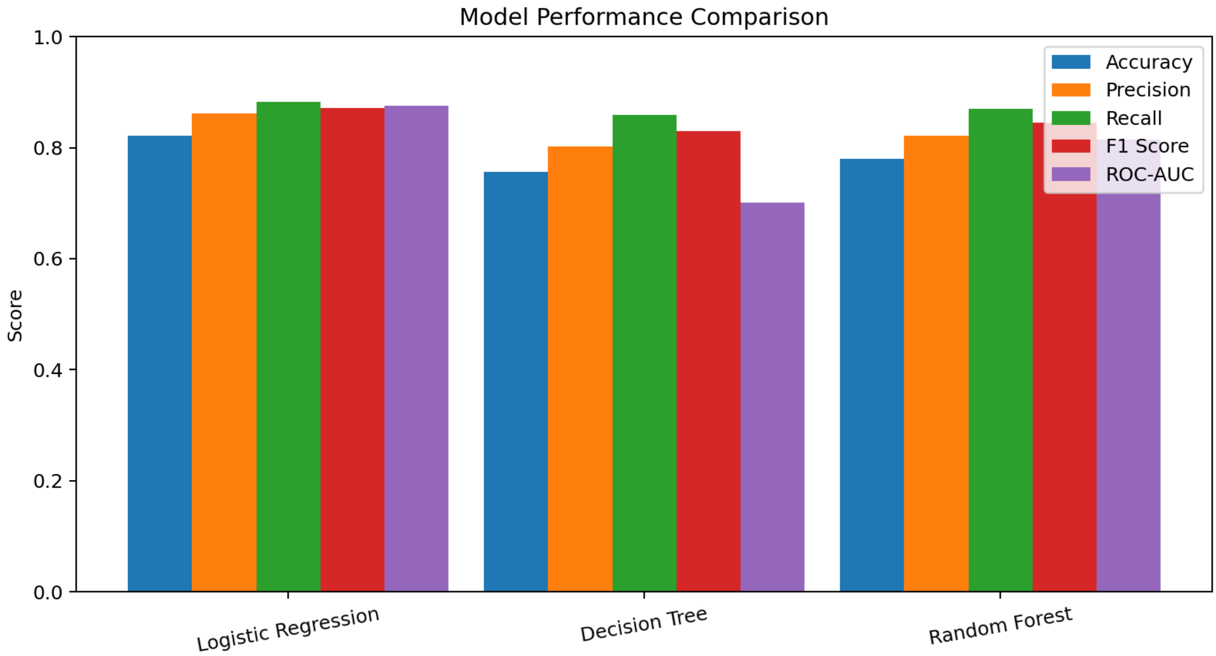

Figure 1: Comparison of classification model performance

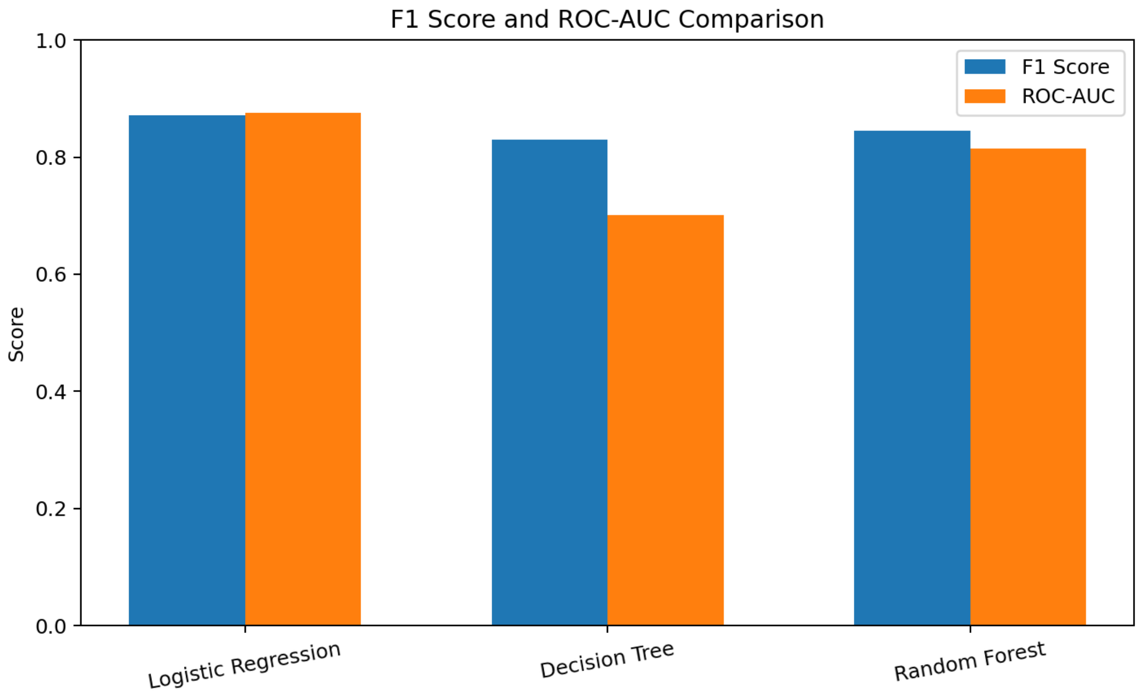

Figure 2: F1-score and ROC-AUC comparison

9. Model Comparison

Logistic Regression achieved the best overall performance with 82.11% accuracy, 86.21% precision, 88.24% recall, 87.21% F1-score, and 87.59% ROC-AUC. Random Forest performed better than Decision Tree, while Decision Tree had the lowest ROC-AUC of 70.05%.

10. Threshold Analysis

Thresholds of 0.30, 0.40, 0.50, 0.60, and 0.70 were evaluated to understand the trade-off between precision and recall. The recommended deployment threshold should be selected based on the best F1-score.

Recommended threshold: [ENTER THE VALUE FROM YOUR COLAB OUTPUT]

11. Business Interpretation

The model can support loan officers by estimating the probability of approval. A lower threshold may approve more applications but can increase incorrect approvals. A higher threshold provides a more conservative strategy but may reject potentially eligible applicants. The final threshold should reflect business risk tolerance and the costs of false approvals and false rejections. Human review should remain part of the final lending decision.

12. Feature Interpretation

Logistic Regression coefficients can identify features with stronger associations with the predicted outcome. Positive coefficients generally push predictions toward approval and negative coefficients toward rejection. These are model associations, not proof of causation.

13. Limitations

- The dataset is relatively small.
- Performance may change on a different population.
- Historical approval patterns may contain bias.
- Available features do not capture every real-world lending factor.
- Human review and lending policies remain important.

14. Conclusion

The project successfully developed a supervised machine learning solution for loan approval prediction. The workflow included preprocessing, SMOTE, model training, comparison, evaluation, and threshold analysis. Logistic Regression was selected as the final model because it achieved the strongest overall results, including 82.11% accuracy and 87.59% ROC-AUC.

15. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, imbalanced-learn (SMOTE), and Google Colab.

16. Deliverables

The deliverables include the Google Colab notebook containing the preprocessing and modeling pipeline, evaluation metrics, charts, threshold analysis, and business interpretation, together with this internship report.